

Chi-Square Test of Independence Analysis

Student Report

Course: Statistical Data Analysis

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Introduction

This report presents a Chi-Square Test of Independence examining whether gender is associated with product preference.

Hypotheses

H0: Gender and product preference are independent.

H1: Gender and product preference are associated.

Observed Frequency Table

Gender	Product A	Product B
Female	2	4
Male	4	2

Expected Frequency Table

Gender	Product A	Product B
Female	3	3
Male	3	3

Results

Chi-Square (manual) = 1.33

Chi-Square (automated) = 1.33

Degrees of freedom = 1

p-value = 0.2482

Conclusion

There is no statistically significant association between gender and product preference. The null hypothesis is not rejected at $\alpha = 0.05$.